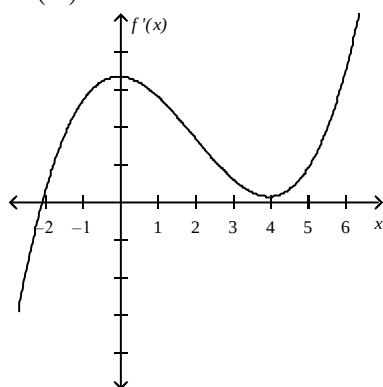


Multiple Choice: Write the CAPITAL LETTER corresponding to the correct answer on the line provided

_____ 1. Below is the graph of $f'(x)$. For what value(s) of x does $f(x)$ have a local minimum?



- A. $x = -2$ B. $x = 0$ C. $x = 4$ D. None

_____ 2. Let $f(x)$ be a continuous function. If $f'(x) > 0$ for $x > c$ and $f'(x) < 0$ for $x < c$ then what type of critical point is $(c, f(c))$?

- A. Local maximum C. Neither a local max nor local min
B. Local minimum D. Unknown

_____ 3. For what values of x is $f(x) = x + \frac{1}{x}$ decreasing?

- A. $-1 < x < 1$ C. $-1 < x < 0, 0 < x < 1$
B. $x < -1$ D. all real numbers

_____ 4. If $f(x) = -2x^3 - 6x^2 + 8$, then which of the following statements is **true**?

- I. $f'(x) > 0$ for $x \in (-2, 0)$
II. $f'(-2) = f'(0) = 0$
III. $f(x)$ has one point of inflection

- A. I only B. I and II only C. III only D. II and III only E. I, II & III

_____ 5. A critical number is:

- A. A number c in the domain of a function f such that $f'(c) > 0$.
B. A number c in the domain of a function f such that $f'(c) < 0$.
C. A number c in the domain of a function f such that $f'(c) = 0$.
D. A number c in the domain of a function f such that $f'(c) = 0$ or does not exist.
E. None of the above

6. True (T) or False (F).

(a) If local max point is $(2, -9)$, then $f'(2) = 0$

(b) A function has an absolute maximum at a if $f(a) \geq f(x)$ for all x in the domain.

(c) A local maximum point can also be an absolute maximum point but not vice-versa.

7. Find the absolute extreme values of the function $g(x) = x^{\frac{2}{3}}(5+x)$, if $-5 \leq x \leq 1$.

8. For Given the function $y = f(x)$ as shown, state

a) the interval(s) where

I. $f'(x) < 0$ & $f(x) > 0$ _____

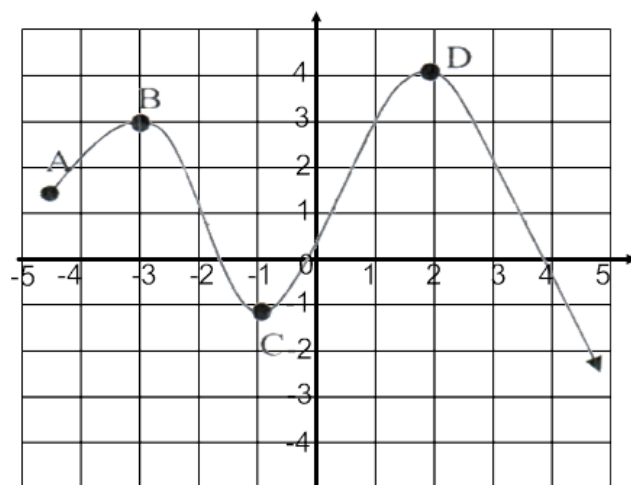
II. $f'(x) > 0$ & $f(x) < 0$ _____

b) the local extrema values (specify the "type")

c) the points where $f'(x) = 0$

d) absolute maximum: _____

e) absolute minimum : _____



9. Find the intervals on which the function $f(x) = (x^2 - 9)^{\frac{2}{3}}$ is increasing and the intervals on which it is decreasing. Determine the x-coordinates of all local extrema.